Venkata Lakshmi Parimala Pasupuleti

Senior Software Engineer · Infrastructure AI · Autonomous AI Agents · LangGraph · Agent Evaluation · DevOps · Python · Kubernetes · Terraform · AWS

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PROFESSIONAL SUMMARY

Senior Software Engineer who designs and builds goal-oriented autonomous AI agents that plan, reason, use tools, and execute multi-step tasks — and the evaluation infrastructure to make them agent-reliable, auditable, and safe in production. LeaseI is an 8-agent LangGraph system with planning and reasoning loops, tool-use orchestration, agent state and memory, error recovery, and human-in-the-loop escalation — every architectural element Airwallex describes for infrastructure AI agents. AgentProbe is a homegrown agent evaluation framework tracking task completion rate, time-to-resolution, escalation rate, hallucination rate, and regression — exactly "build evaluation frameworks to measure whether agents reliably achieve their goals." Infrastructure foundation: Kubernetes, Terraform, deployment orchestration, infrastructure provisioning, Prometheus, Grafana, AWS/GCP, CI/CD, PostgreSQL, schema migrations, OpenTelemetry. 8+ years of production Python engineering. M.S. Data Analytics Engineering, George Mason University (GPA 3.87). — github.com/venkatalakshmiparimala/AgentProbe

WORK EXPERIENCE

Blue Horse Digital Sep 2026 – Present

AI/ML Engineer (Volunteer) · Remote

- Building ML models and autonomous AI pipelines; designing agentic components that interact with backend APIs; debugging production reliability issues; using AI coding agents as core workflow tools.

Magizhs Tech Solution May 2025 – Aug 2025

AI/ML Engineer Intern · Remote

- Designed and implemented a full autonomous AI agent system: planning and reasoning loops (LangGraph), tool-use orchestration, RAG over regulatory corpus, safety guardrails as hard enforcement logic, deployment orchestration of the agent pipeline. Agents autonomously execute multi-step DevOps-style compliance workflows end-to-end with minimal human intervention.

George Mason University Aug 2024 – Aug 2025

Research Associate + Teaching Assistant

- Built multi-source ML prediction pipelines (Python, R) across 190+ countries (R^2=0.977); HuggingFace fine-tuning + inference API deployment; statistical modeling and validation. Workshops to 600+ students across Python and SQL.

Entrata (PropTech / Real Estate SaaS) Sep 2018 – Aug 2024

Senior Software Engineer · Remote

- **Infrastructure, DevOps, and production operations at scale** (Python, Java, Kotlin, PostgreSQL, RabbitMQ, AWS, Kubernetes) — owned 20+ production modules at 99.5% uptime. Infrastructure provisioning and deployment orchestration via Terraform + Kubernetes; schema migrations across hundreds of thousands of records with rollback preparation; Docker microservices redesign (50% scalability improvement); AWS (S3, EC2, RDS, Lambda, Step Functions). Operated as the de-facto owner for production incident resolution through to permanent root cause fix.
- **DevOps pipelines, observability, and database operations** — CI/CD (GitHub Actions) with automated testing and deployment; 200+ PostgreSQL query optimizations (30% gain via execution plan analysis, indexing, partitioning, stored procedure refactoring); database schema design and migrations across multi-tenant systems; OpenTelemetry + Prometheus + Grafana observability (40% debugging time reduction); Python/Bash automation eliminating 80% of manual operational workload. Auditable, PHI-compliant data pipelines (zero security incidents across 7 years).
- **Global cross-functional collaboration and team growth** — collaborated with global engineering teams in India; 300+ code reviews; mentored 5+ engineers; 120+ Agile sprints; engaged cross-functional stakeholders to translate operational pain points into technical solutions.

PROJECTS

AgentProbe: Agent Evaluation Framework & Reliability Infrastructure TASK COMPLETION METRICS CI REGRESSION

Python · FastAPI · LangGraph · ChromaDB · GitHub Actions · OpenTelemetry | github.com/venkatalakshmiparimala/AgentProbe

- Homegrown agent evaluation framework for measuring **agent reliability** in production: **task completion rate**, time-to-resolution, escalation rate, hallucination rate, groundedness, and latency; **CI regression harness** with golden datasets so agent behavior doesn't silently degrade; LLM-as-judge classifiers; **auditable AI** trace-level observability; multi-provider LLM routing with cost controls; offline + online evaluation loops. Built to ensure autonomous AI agents are safe, auditable, and predictable — the reliability standard infrastructure agents must meet.

Leasel: Goal-Oriented Autonomous 8-Agent System LANGGRAPH HITL + GUARDRAILS

Python · LangGraph · FastAPI · ChromaDB · MCP · SQLite · OpenAI API · LangSmith | github.com/venkatalakshmiparimala/Lease_complaince_rag

- Production autonomous AI agent system: **planning and reasoning loops** (8-step: Triage → Retrieval → Configuration → Historical → Reasoning → Validation → Escalation); **tool-use orchestration** (external APIs, document retrieval, structured data queries); **agent state and memory** (durable state, ChromaDB knowledge store, contextual retrieval); **error recovery** and **human-in-the-loop escalation** when confidence falls below threshold; **safety guardrails** routing regulated outcomes to humans. 93.6% precision, 87.9% F1 across 1,240 test cases — evaluated with AgentProbe.

EDUCATION

M.S. Data Analytics Engineering George Mason University, Virginia May 2026

GPA: 3.87

B.Tech Computer Science & Engineering JNTU Anantapur 2017

SKILLS

Agentic AI & Evaluation: LangGraph, LangChain, LangSmith, MCP, Autonomous AI Agents, Goal-Oriented Agents, Planning & Reasoning Loops, Tool-Use Orchestration, Agent State & Memory, HITL Escalation, Safety Guardrails, Error Recovery, Agent Reliability, Auditable AI, Agent Evaluation Frameworks, Task Completion Metrics, Time-to-Resolution, Escalation Rate, CI Regression Harnesses, LLM-as-Judge, Hallucination Detection, Trace-Level Observability, Multi-Provider LLM Routing, OpenAI API, Anthropic API, RAG Pipelines, ChromaDB, Prompt Injection Defense

Infrastructure, DevOps & Cloud: Kubernetes (K8s), Terraform, Infrastructure Provisioning, Deployment Orchestration, DevOps, AWS (S3, EC2, RDS, Lambda, Step Functions, Bedrock), Google Cloud Platform, Docker, Microservices, Distributed Systems, CI/CD (GitHub Actions), System Design, Fault-Tolerant Systems, Idempotency, Schema Migrations, Multi-Tenant Architecture, PHI Compliance, HIPAA, SOC 2

Observability & Data: OpenTelemetry, Prometheus, Grafana, Amazon OpenSearch, Elasticsearch, PostgreSQL (8+ yrs, query optimization, execution plan analysis), MySQL, Redis, RabbitMQ, Apache Kafka, Python (8+ yrs), Java (Spring Boot), Kotlin, TypeScript, Node.js, FastAPI, Performance Engineering, Agile/Scrum